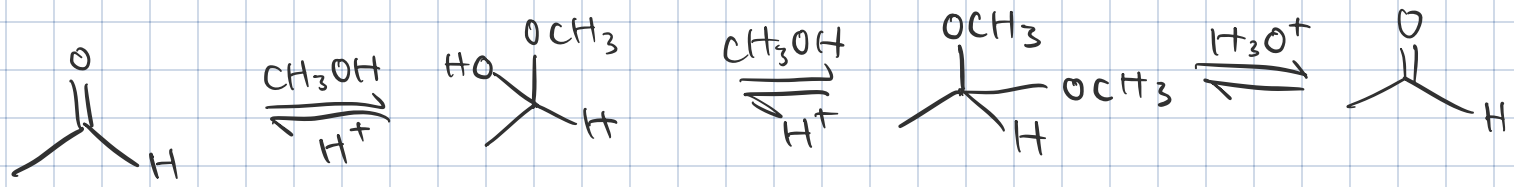
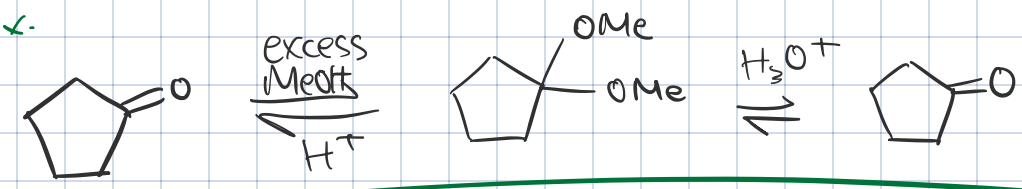
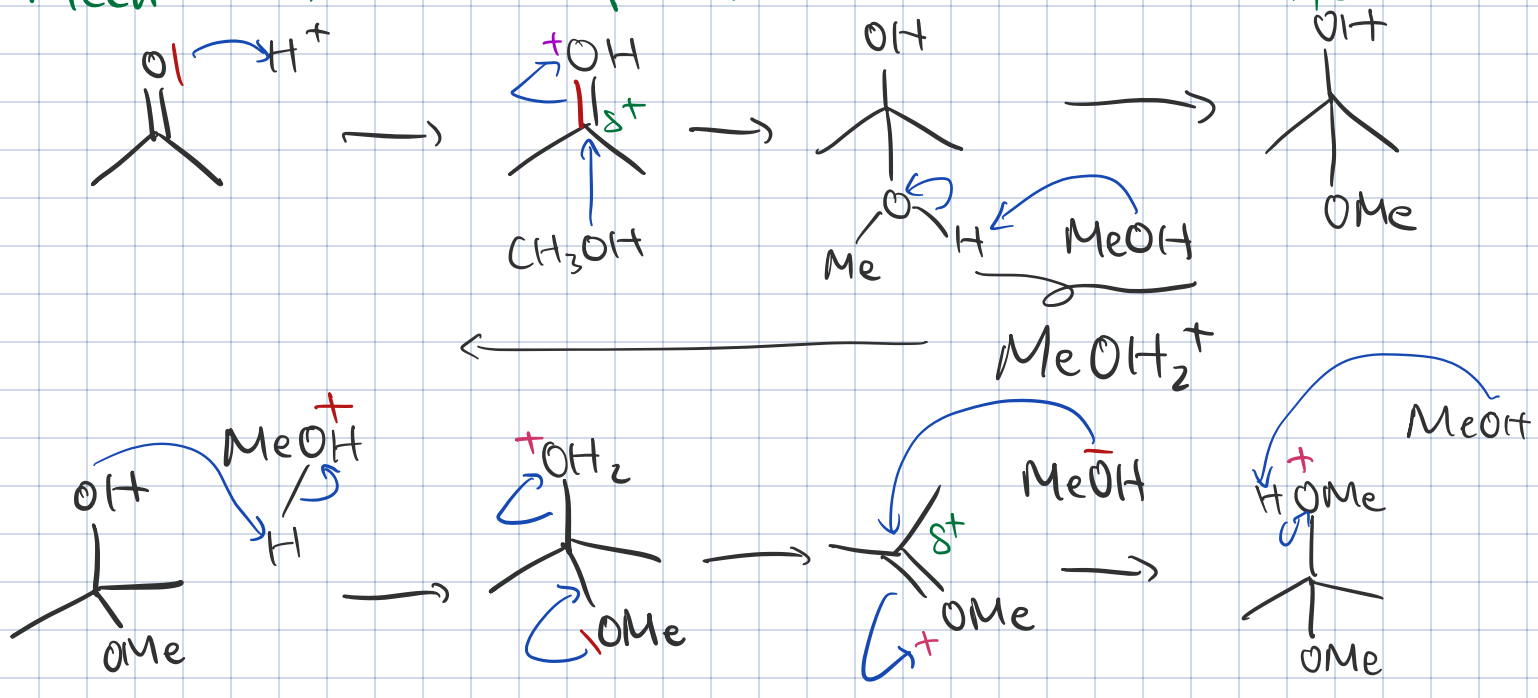


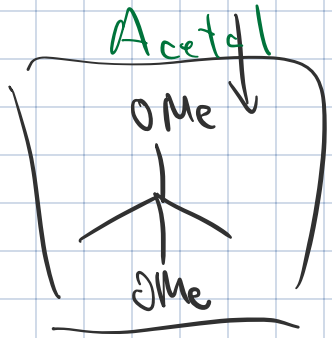
Ex.



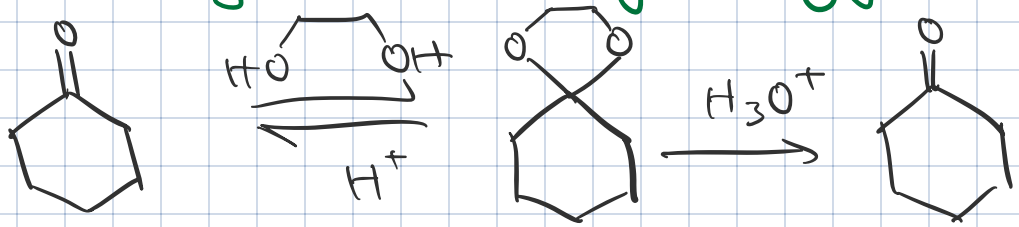
Mech for acetal formation



Note how 2 ROH's were added, 1st to make Hemi, 2nd to make Acetal

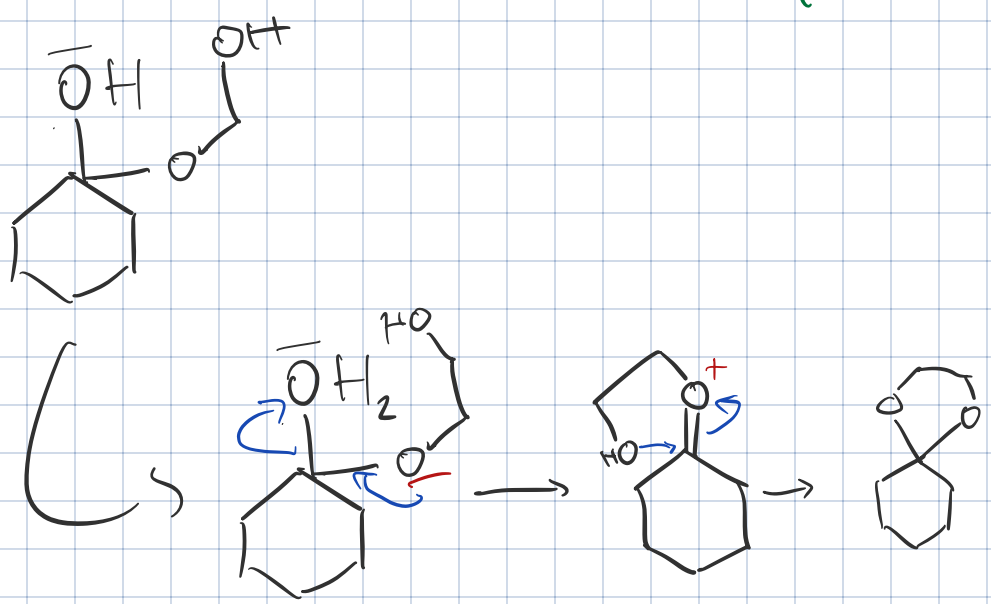


Protecting group Ethylene glycol

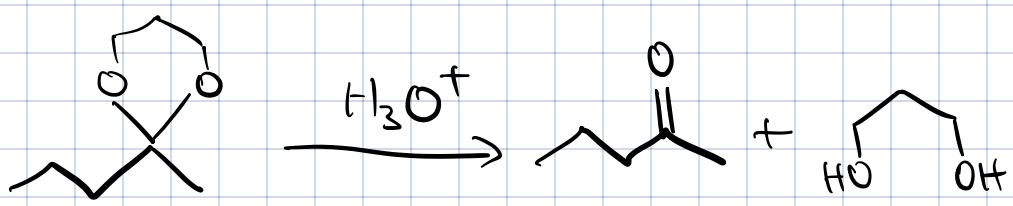


Protection from reducing agents

Hemi

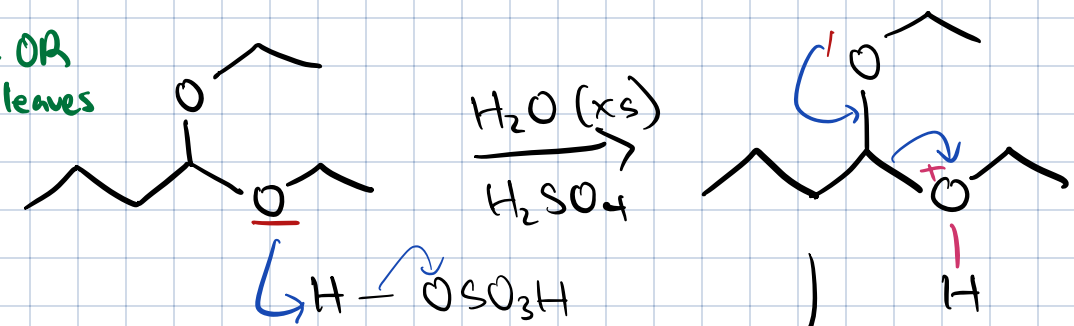


Acetal hydrolysis



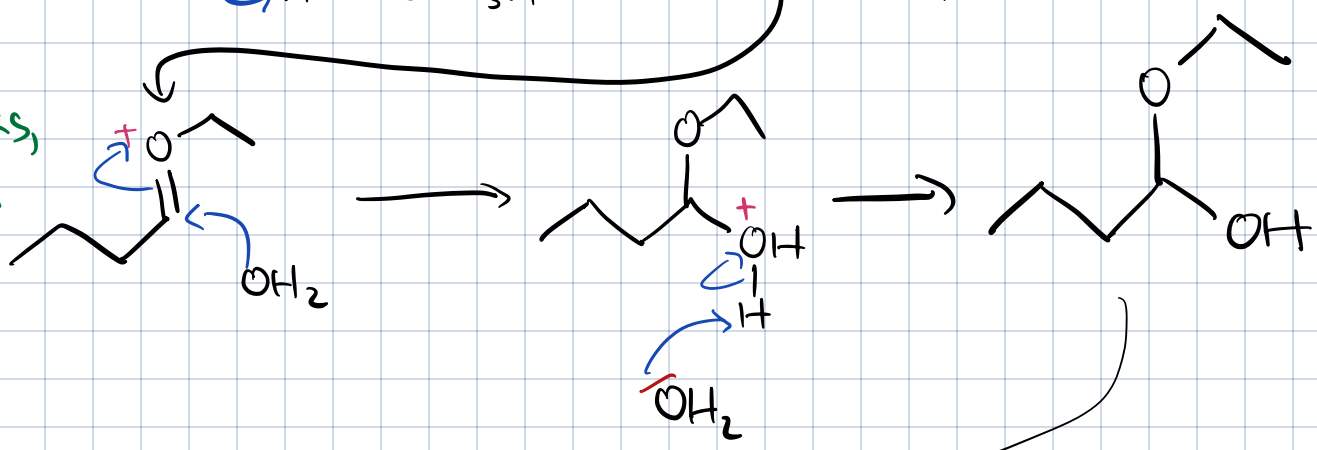
Mech

1. Protonate OR and $^{+}$OR leaves

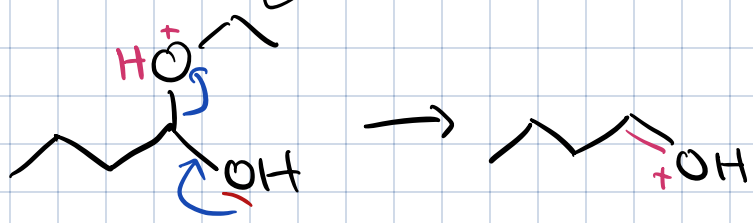


In excess

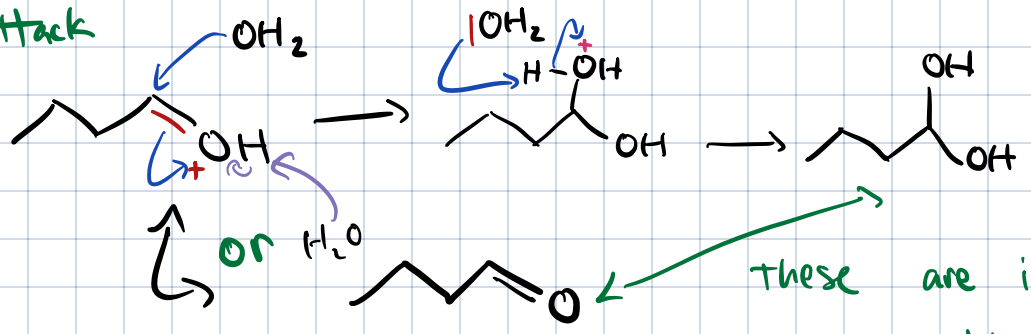
2. H_2O attacks, hemiacetal formed



3. Protonate 2nd OH group & it leaves

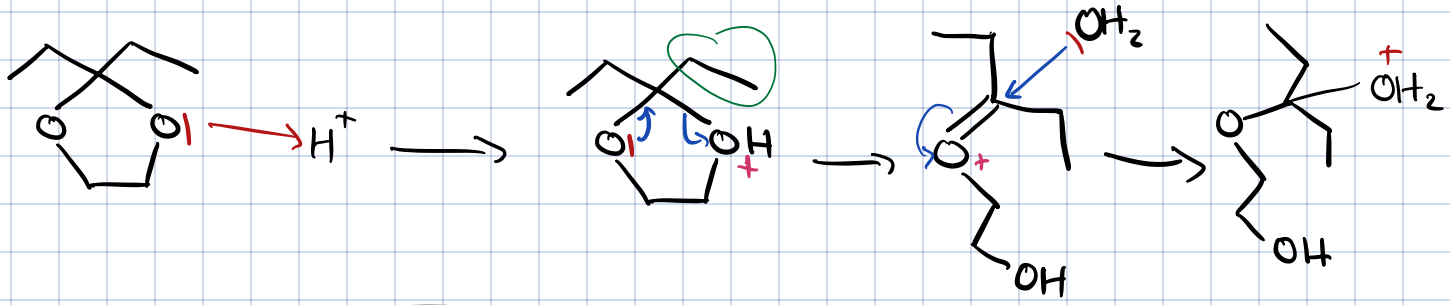


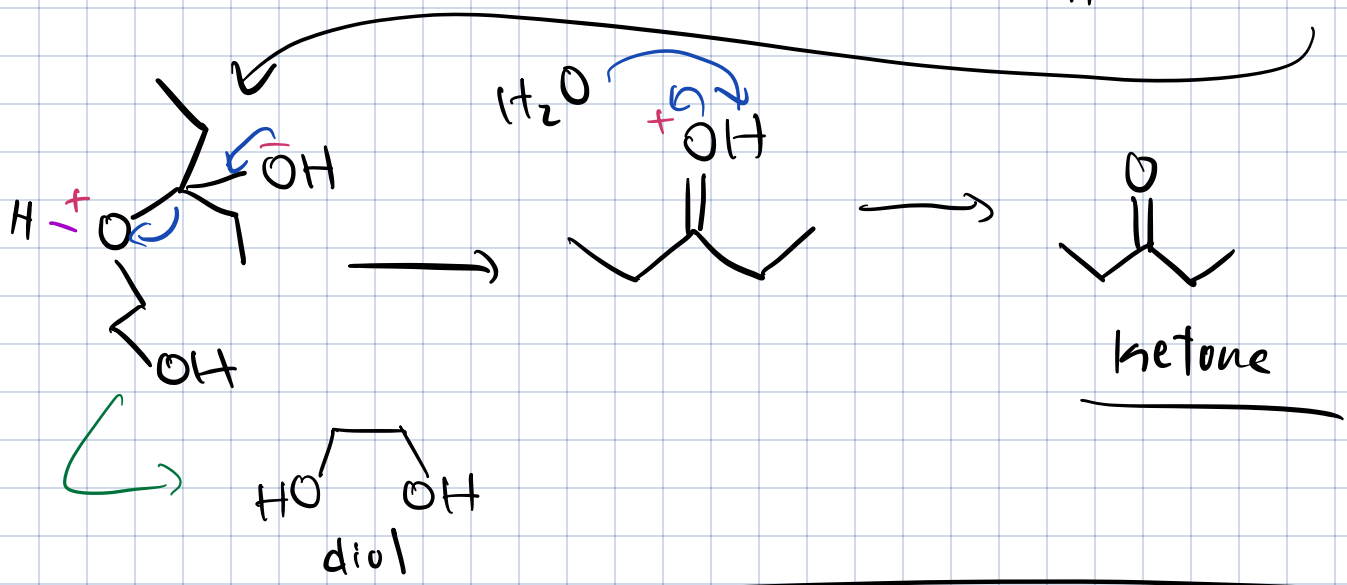
4. H_2O attack



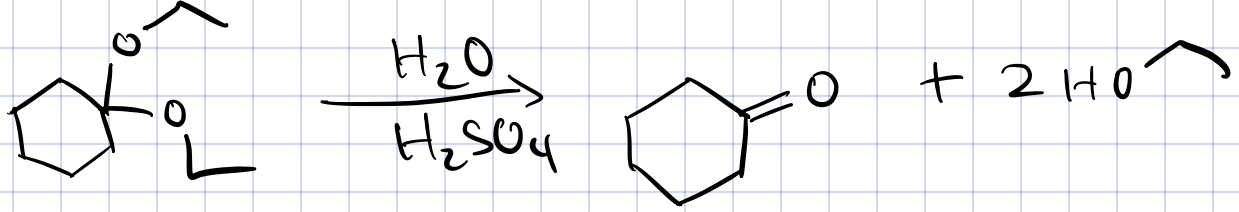
These are in eq in water "hydration of ketones/aldehydes"

Cyclic Example





Trick


O
O
H_2O
H_2SO_4
O
+ 2 HO